Mateus Alves da Rocha

Brasília, Brazil

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Skills

Languages: C++, Python, Shell
Backend & Systems: Distributed Systems, Performance Optimization, System Design
Cloud & Platforms: AWS (EKS, EC2, S3, ECR), Kubernetes, Linux
Storage & Protocols: Aerospike, LMDB, Protobuf, HTTP, MQTT
Build & Testing: Bazel, CMake, Jenkins, CI/CD, Git, GTest
Observability & Debugging: Grafana, Prometheus, GDB, Profiling, Root Cause Analysis

Work Experience

Sigma Software Group - USA Client - Brazil (Remote) — Aug. 2024 – Present

Software Engineer – Backend Systems

- Develop and maintain performance-critical C++ and Python backend services for a high-throughput real-time advertising platform deployed on AWS using Kubernetes.
- Reduced execution time in a production data-access path from approximately 650 ms to 150 ms (77%) by removing redundant application-side sorting after verifying database indexes already guaranteed the required ordering.
- Designed an LMDB-backed local storage architecture for a 37M+ record dataset to replace gigabyte-scale Protobuf snapshots loaded entirely into memory, addressing pod OOM failures and targeting approximately 1 GB lower memory usage per service instance.
- Led backend development of Pause Ads by extending the existing real-time bidding pipeline to support programmatic bidding on pause-screen inventory while preserving existing video behavior.
- Diagnose and resolve production issues in distributed backend services using logs, metrics, Grafana, and Prometheus, identifying root causes and implementing reliability fixes.

Onboard Mobility - Brazil (Remote) — Aug. 2023 – Aug. 2024

Software Engineer

- Reduced denylist lookup complexity from O(n) to average O(1) by replacing linear scans with a hash-based C++ implementation, validating the improvement through profiling.
- Introduced unit testing with Google Test across C++ components that lacked automated coverage, improving regression safety and enabling safer refactoring.

APTIV - Poland (Hybrid) — Mar. 2022 – Jul. 2023

Embedded Systems Test Engineer

- Built an internal web application integrated with the Jira API to replace an ad-hoc cross-team request workflow, achieving adoption by 5 teams within the first two weeks of launch.
- Built Jenkins-based automation pipelines in Python for repetitive testing workflows, reducing manual effort and improving execution consistency.

Onboard Mobility - Brazil (Remote) — Oct. 2019 – Mar. 2022

Embedded Software Engineer – Earlier Tenure

- Developed C++ remote device management capabilities over MQTT, enabling diagnostics, health monitoring, and remote command execution across distributed field devices.
- Redesigned device-to-cloud communication by replacing HTTP-based transfers with encrypted MQTT messaging to an AWS backend, reducing bandwidth usage and improving communication efficiency.

Kinology - Brazil (Onsite) — Jun. 2018 – Oct. 2019

Hardware/Firmware Development Engineer

- Redesigned an embedded product architecture by migrating from an ATtiny84 plus Bluetooth module to an ESP32-based solution, reducing hardware cost and simplifying the overall system design.

Education

Bachelor of Electronics Engineering — Dec. 2018

University of Brasilia, DF - Brazil

International Exchange program by Ciências Sem Fronteiras — Aug. 2015 – Aug. 2016

Wayne State University, MI / University of California, Los Angeles (UCLA), CA - USA